GENERAL GUIDELINES FOR FITTING WORK

In this modern era, automatic machines are used for manufacturing in small, medium and heavy industries. But fitting and bench work finds an important place in completing and finishing a job to the desired accuracy. Some components that come out after machining processes require some minor operations to be performed by hand tools. Fitting work is an important method in doing this. Fitting is the assembling of parts together and removing them for necessary fit. Both these types of work require many number of hand tools, devices and equipments.

Fitting of parts are made with different kinds of fits. It may be arranged where the fit is tight, loose or very loose. The parts may be arranged to move together with no relative movement or rotational and sliding relative movement.

OBSERVATION EXERCISE FOR STUDENTS

We happen to see various machine tools in engineering shops. We see the sliding and rotating parts are made to fit into each other by Dovetail joints, T – joints, L – joints triangular joints and as holes and shafts. These joints should be closely watched to understand their nature and their respective uses.

For example, in a lathe – the parts of the assembly of carriage – saddle, cross – slide and compound slide. Observe these joints for their shape and purpose.

Observe the top of the table of a shaping machine – What is the shape of the slots made on it? What is its purpose ?

Again in the shaping machine, the ram reciprocates on the column. Observe the joint between the top of the column and bottom of the ram for their shape and purpose.

Observe the method of joining of the parts column, Knee, Saddle and table of a milling machine. These parts are joined by different methods of joining and for different purposes.

Likewise different parts of different machine tools are joined by different methods for obtaining different utilities.

We can also observe some house – hold articles are also joined by different methods.

1. The cap and the body of the bottles that we use daily. How are they joined ?
2. The tiffin – carrier. How are the set of dishes held together ?
3. Hot – cases. How is the lid joined with the base ?
4. The base vessel and the lid of the pressure cooker. What are the methods by which they are joined ?Why ?
5. The regulator of the L.P.G cylinder. How is it fitted with the cylinder ?
6. The frames and the doors of the windows. How are they fitted ?